Multimodal Emotion Recognition

Detecting Emotions in Travel Twitter Posts

NLP | COMPUTER VISION | DEEP LEARNING | INTELLIGENT SYSTEMS

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Why Is This Hard?

Image Only — Misleading

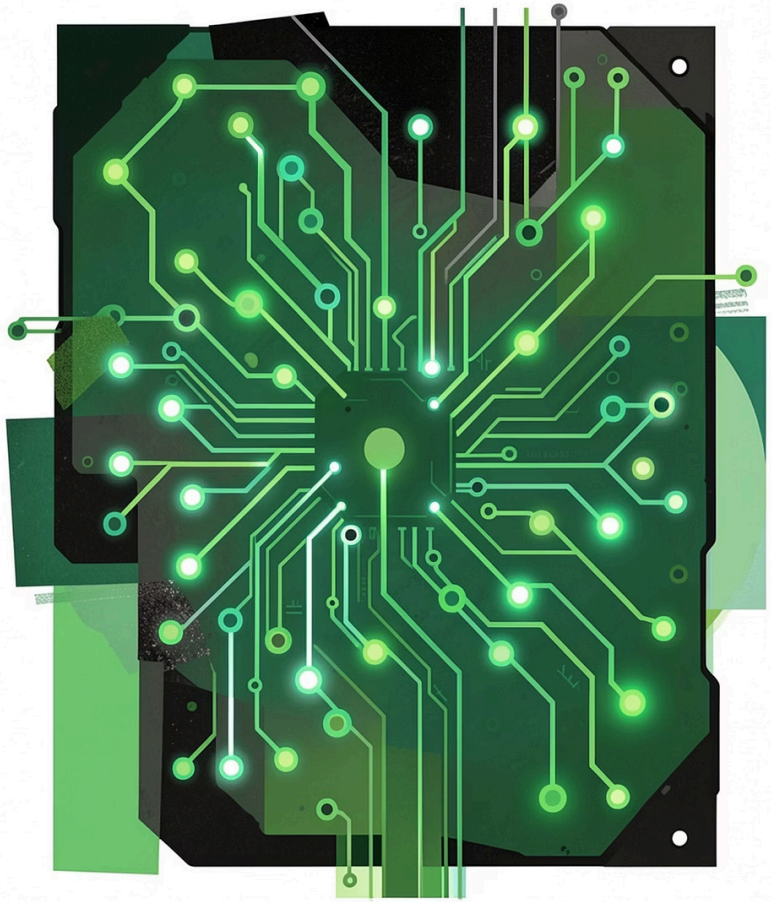
A beach photo looks positive – but the caption reads *"worst trip ever."* Pixels lie without context.

Text Only — Ambiguous

"sick view" is positive slang – classical NLP scores it as **negative**. Semantics shift with culture.

Solution: Image + Text combined

- **F1 = 0.76** vs random baseline 0.33



5 Modules — 5 Subjects

CV: ResNet18 Transfer
Transfer learning
F1=0.39

Input
Twitter photo + tweet

CV: HSV + SVM
Classic vision F1=0.37

Late Fusion Emotion Classifier

NLP: TF-IDF + LR
Classic text model
F1=0.60

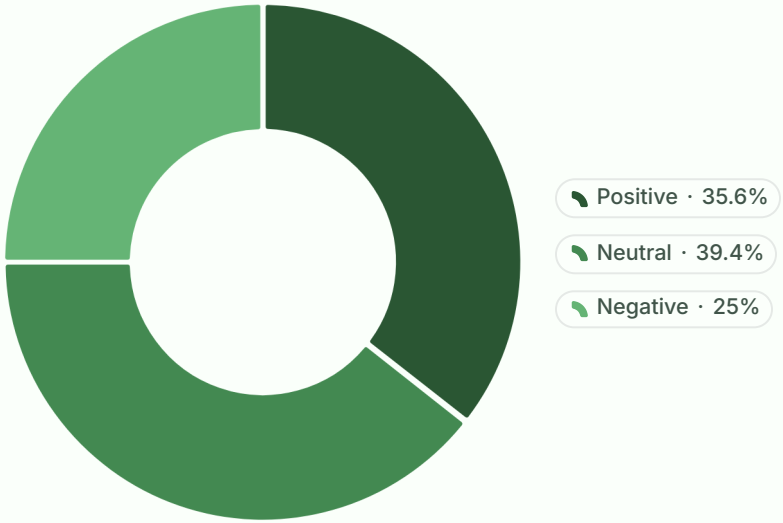
NLP: BERT Fine-tuned
Fine-tuned text
F1=0.74

NLP: RoBERTa Zero-shot
Zero-shot text
F1=0.67

The **Late Fusion Agent** uses Simulated Annealing to find optimal weights: LR=0.36 · BERT=0.38 · SVM=0.00 · ResNet=0.15 · RoBERTa=0.11, maximising macro-F1 on the validation set.

Final output: positive / negative / neutral – F1 = 0.76

MVSA-Single Dataset



Metric: **Macro-averaged F1** – chosen to handle class imbalance fairly across all three labels.

4,869

Twitter Pairs

Image + caption per sample

3

Emotion Classes

Positive · Neutral · Negative

13.1

Mean Caption Length

Words per tweet

Step-by-step Workflow

Data
4,869 posts · 80/20 stratified split

Preprocessing
Text: lowercase → URLs → stemming · Image: 224×224 · ImageNet norm

Training
Text → CPU (Apple M1) · BERT + ResNet18 → Colab T4 GPU

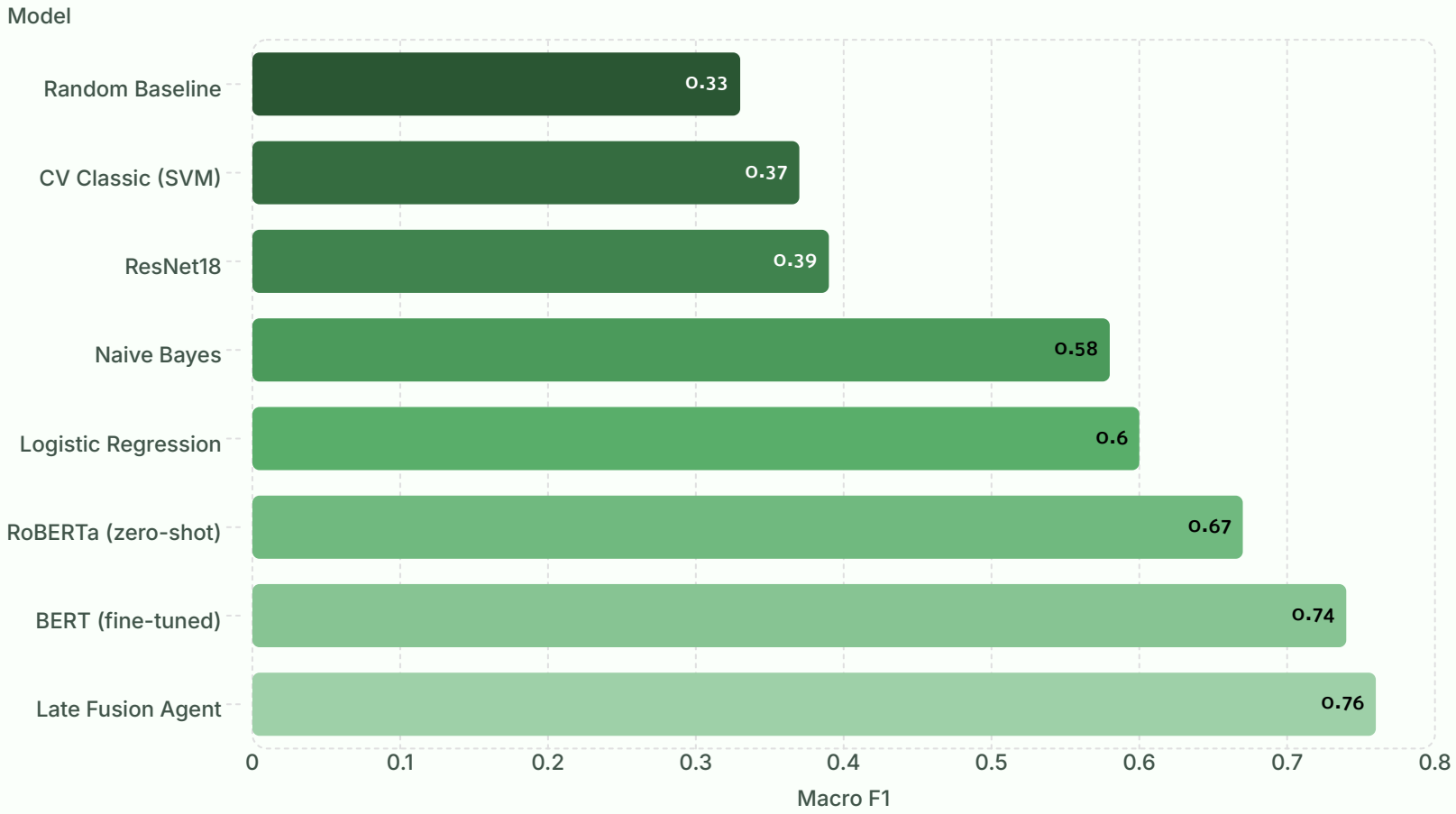
Probability Output
Each module → P(pos), P(neg), P(neu) · 5 vectors of size 3

Fusion Optimisation
Simulated Annealing · 135 iterations · ~2 min convergence

Prediction
argmax of weighted average → **F1 = 0.76**

Same random_state=42 split across all modules – ensures fair comparison

Fusion Beats Every Module



Key insight: **Text >> Image** – **35-point gap** between best text (0.74) and best vision (0.39) model, Fusion Wins, Zero-shot Power.

The System Explains Itself

Grad-CAM — Where ResNet18 Looks

Negative

Activates on deterioration regions – damage, crowds, dirt.

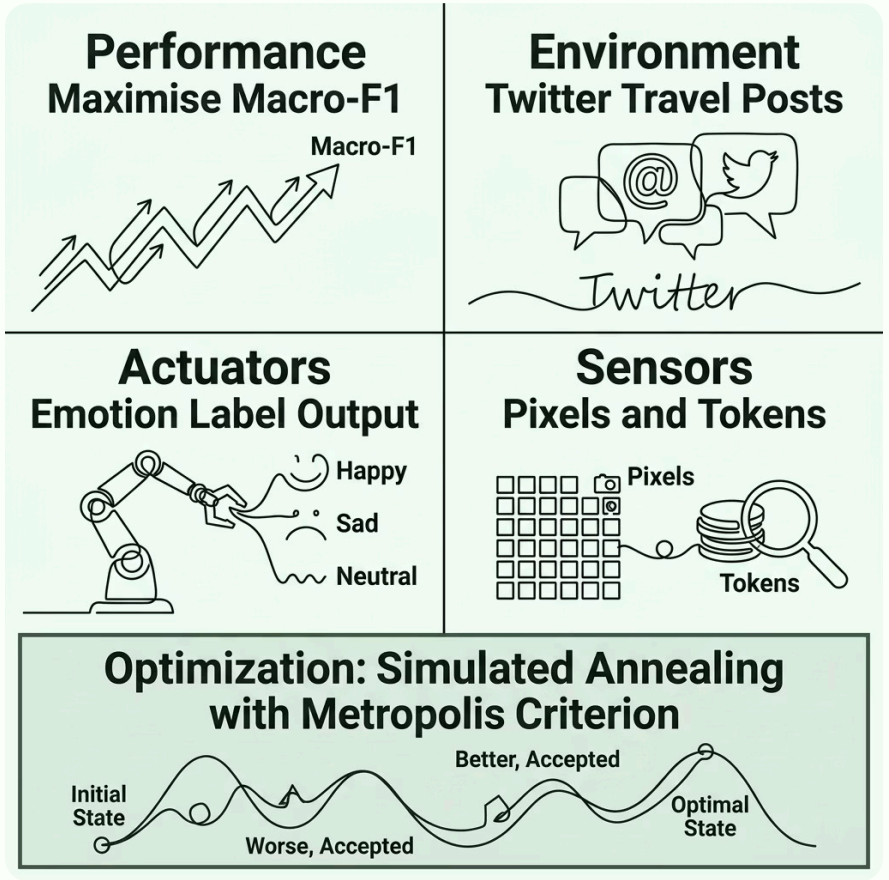
Neutral

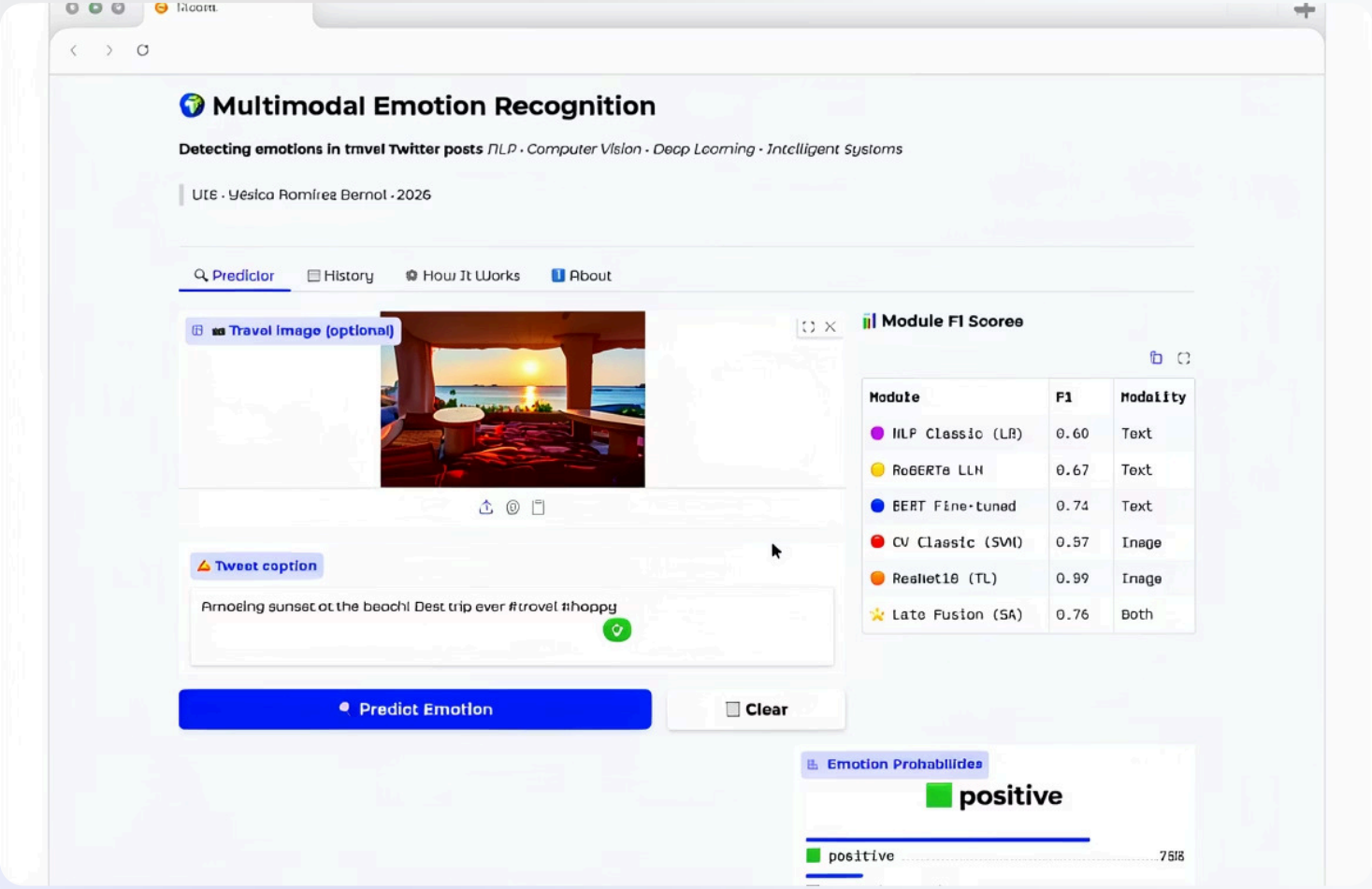
Activates on embedded text within the image.

Positive

Activates on scenic vistas and relaxation areas.

PEAS Framework — Rational Agent





Live Demo

Gradio Interactive Interface

Real-Time Prediction

All 5 modules run in parallel on any image-text input.

Per-Module Breakdown

Confidence scores displayed for each model individually.

Grad-CAM Map

Visual attention overlay shows where ResNet18 focused.

Session History

Previous predictions logged for review and comparison.

What Did We Prove?

1

Fusion Wins

Multimodal Late Fusion **F1 = 0.76** outperforms every individual module – the whole exceeds the sum of its parts.

2

Text Dominates

Tweets carry far more emotional signal than travel photos on Twitter

3

Pretraining Works

RoBERTa zero-shot surpasses trained classical NLP – domain pretraining is a decisive advantage.

Future work: Cross-modal attention · Conflict detection between modalities · RoBERTa fine-tuning on MVSA

THANKS.